



Project

Impact of Quantum Computing on Elliptic Curve Cryptography (ECC)

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Table of Contents

1.0 Introduction	1
1.1 Overview of Elliptic Curve Cryptography (ECC)	1
1.2 Elliptic Curve Discrete Logarithm Problem (ECDLP)	2
2.0 Background Study	3
2.1 Shor's Algorithm	3
2.2 Post-Quantum Cryptography (PQC)	4
2.2.1 Lattice-Based Cryptography	5
2.2.2 Code-Based Cryptography	5
2.2.3 Hash-Based Cryptography	5
2.2.5 Isogeny-Based Cryptography	6
2.3 Hybrid ECC-PQC Approaches	6
3.0 Literature Review	7
3.1 Paper 1: Hardware Deployment of Hybrid PQC (Azarderakhsh et al., 2021)	7
3.2 Paper 2: Post Quantum Cryptography Review (Bavdekar et al., 2022)	8
3.3 Paper 3: Post-Quantum Readiness and Migration (Gupta & Mittal, 2026)	8
3.4 Paper 4: Cryptographic Dependency Inventory (Lee et al., 2026)	8
3.5 Paper 5: Hybrid PQC Authentication for 6G (Turnip et al., 2026)	9
4.0 Critical Discussion	9
4.1 The Quantum Threat to ECC-Based System	9
4.2 Current Challenges in Replacing ECC with PQC	10
4.3 Advantages and Disadvantages of Hybrid ECC-PQC	10
4.4 Current Research Trends and Future Directions	11
4.4 Critical Evaluation and Recommendations	11
5.0 Conclusion and Future Work	11
5.1 Summary of Key Findings	11
5.2 Future Research Directions	12

1.0 Introduction

Elliptic Curve Cryptography (ECC) has become a cornerstone of contemporary public-key infrastructure, finding extensive deployment across numerous digital security domains. Its applications span secure web communications via TLS/SSL protocols, digital signature schemes, cryptocurrency platforms, mobile device security, and Internet of Things (IoT) ecosystems. The efficiency of ECC especially when it comes to ability to achieve high level of security with smaller key sizes which makes it secure compare to other methods. With the growing adoption of digital means of communication, ECC plays an increasingly significant role in providing data security against cyber-attacks.

However, current encryption techniques face certain challenges due to developments in quantum computing. ECC and other classical cryptography rely on issues that are difficult to solve on a traditional computer. These issues can now be resolved with quantum computers thanks to the development of quantum algorithms like Shor's algorithm. Hence, it is essential to examine ECC mechanism, assumptions about its security, as well as the influence of quantum computing on its security in the future. This chapter provides an overview of ECC concept, ECDLP, and basic principles of quantum computing.

1.1 Overview of Elliptic Curve Cryptography (ECC)

The Elliptic Curve Cryptography (ECC) is a type of cryptography which uses elliptic curves over finite fields to provide implementation for public key cryptography schemes. ECC does not use the hard problem of factoring large integers (as done by RSA algorithm) but rather uses the mathematics of elliptic curves. An elliptic curve can be described using the following equation:

$$y^2=x^3+ax+b$$

In this case, a and b are constants which should meet certain criteria so that the curve does not have any singularities. The points that lie on the curve along with a special operation referred to as addition form a mathematical group. In the ECC, all the crypto functions employ the multiplication of the point by itself. This process is computationally very simple to execute but very complex to reverse. Among the key characteristics of ECC are:

- **Short keys:** Using short keys, ECC offers a comparable level of security (256-bit ECC equals 3072-bit RSA).
- **Efficiency:** ECC is a suitable option for IoT and mobile applications because it uses less energy and processing power.

- **Security:** Security based on hard mathematical problems (ECDLP). ECC is widely utilized in TLS, SSH, blockchain systems, and digital signature algorithms like ECDSA because of the aforementioned factors.

SYMMETRIC KEY LENGTH (BIT)	RSA KEY LENGTH (BIT)	ECC KEY LENGTH (BIT)
80	1024	160
112	2048	224
128	3072	256
192	7680	384
256	15360	521

*Figure 1: Comparison of Key Sizes for Symmetric, RSA, and ECC Cryptosystems
(GlobalSign, 2025)*

1.2 Elliptic Curve Discrete Logarithm Problem (ECDLP)

The difficulty of solving the Elliptic Curve Discrete Logarithm Problem (ECDLP) determines the security of ECC. The definition of ECDLP is as follows - given two elliptic curve points, PPP and QQQ, were:

$$Q = kP \quad \text{or} \quad P = kQ$$

The task here is to obtain the integer kkk which is called the private key.

It is difficult to extract kkk from PPP and QQQ, although it is easy to calculate QQQ from PPP and kkk (one way computation). If key sizes are sufficiently large, the problem is computationally challenging for traditional algorithms to solve.

ECDLP characteristics:

- For sufficiently big keys, there is currently no effective classical solution for solving the ECDLP problem.
- Even the most effective classical methods (like Pollard's rho algorithm) still require exponential amounts of time.
- The security of the ECC-based system is guaranteed because the ECDLP is difficult.

However, if quantum computing technology is used, the scheme's security is compromised. In fact, Shor's method suggests that a quantum computer may effectively tackle the problem.

1.3 Basics of Quantum Computing

Quantum Computing is an innovation that brings about a new paradigm in information processing based on principles of quantum mechanics. While conventional computers use bits of either one state or the other, quantum computers make use of qubits that can be in more than one state due to the principle of superposition. The following are the fundamental ideas underlying quantum computation:

- **Superposition:** The qubit can hold both 0 and 1 at the same time.
- **Entanglement:** Qubits' states can be coupled so that, regardless of distance, one qubit's state influences another.
- **Interference:** It is possible to alter quantum states so that incorrect answers are muted and correct replies are magnified.

When it comes to addressing issues, quantum computers are superior to traditional ones because of all these properties. The creation of Shor's algorithm, which effectively solves the discrete logarithm and integer factorization problems, is one of the most significant accomplishments. The implications for public key cryptography are dire. RSA which makes use of factorization, is easily cracked. ECC which makes use of ECDLP, is also cracked. The rapid development of quantum computing suggests that such devices will be feasible in the future, even though large-scale and fault-tolerant quantum computers are not yet common. This has led to an increase in interest in post-quantum cryptography (PQC).

2.0 Background Study

2.1 Shor's Algorithm

Shor's Algorithm was first proposed in 1994 by Peter Shor; it was formally published in 1997 as part of a threat to the long-term viability of Elliptic Curve Cryptography (ECC) with a quantum computer. It allows the use of a quantum computer to solve both the Integer Factorization problem and the Discrete Logarithm problem in polynomial time. Both of these problems are difficult to solve for classical computers (Shor, 1997).

The security of ECC relies on the Elliptic Curve Discrete Logarithm Problem (ECDLP); because obtaining a public key from a private key is an easy task, however, reversing the above is assumed to be difficult or impractically difficult to do using classical computing techniques. The ability to perform the reverse is based upon the fact that there is no known efficient way to compute a discrete logarithm over an elliptic curve. Therefore, ECDLP provides the foundation for the security of ECC. Unfortunately, Shor's Algorithm fundamentally alters the belief about the hardness of ECDLP. By providing an adversary sufficient computational power, such as a sufficiently large number of qubits and low error rates, Shor's Algorithm enables an adversary to recover the private key given the corresponding public key much faster than would be possible with any classical computer (Shor, 1997).

One of the primary reasons that Shor's Algorithm works well is due to its ability to take advantage of quantum mechanical properties of particles. Specifically, Shor's Algorithm uses three quantum mechanical phenomena in order to evaluate multiple computational possibilities

at once. These include superposition, entanglement and the Quantum Fourier Transform (QFT). Due to the ability to evaluate multiple computational paths at once, Shor's Algorithm greatly reduces the amount of time needed to solve ECDLP. As a result of this dramatic increase in speed, currently implemented ECC will be insecure once fault tolerant quantum computers have been developed (Shor, 1997).

While Shor's Algorithm specifically addresses the security of ECC, other commonly used public-key cryptographic systems are similarly threatened. These include RSA, Diffie-Hellman and DSA. All of these public-key cryptographic systems rely upon mathematical problems that can be solved by a sufficiently powerful quantum computer running Shor's Algorithm. Therefore, if Shor's Algorithm becomes feasible in terms of execution time, then a wide variety of secure communication mechanisms, including digital signatures, cryptocurrencies and key exchange mechanisms will be compromised (Shor, 1997).

In addition to posing a direct threat to the integrity of public-key cryptography systems, Shor's Algorithm also poses a significant indirect threat. That is, a malicious user may begin collecting encrypted data now with the expectation that they will eventually have access to a suitably powerful quantum computer when they can decrypt previously unreadable data. Because users cannot know when their adversaries will gain access to such technology, this creates uncertainty about what constitutes reasonable security practices in light of the existence of Shor's Algorithm. This increased sense of urgency has motivated numerous organizations within industry and government to develop new technologies that will provide sufficient protection against attacks using quantum computers (NIST, 2023).

2.2 Post-Quantum Cryptography (PQC)

Post-Quantum Cryptography (PQC) is cryptographic methods or techniques designed to be resistant to attacks by both conventional and quantum computers. In contrast with Quantum Cryptography, PQC uses no specialized quantum hardware; instead it uses the difficulty of certain computational mathematical problems that are believed at this time to be very difficult to solve using a quantum computer (NIST, 2024). Concerns regarding potential future attacks from quantum computers prompted a surge in research related to PQC during the last ten years. The publication of NIST's first set of standards for post-quantum cryptography in 2024 represented an important milestone toward enabling organizations to start integrating these standards into their long-term plans for developing comprehensive cybersecurity measures (NIST, 2024).

2.2.1 Lattice-Based Cryptography

Lattice-based encryption methods are now considered to be a top method for families of post-quantum cryptographic algorithms. The basis of its security comes from challenging mathematical problems that include Learning with Errors (LWE) and Module Learning With Errors (MLWE), neither of which has an efficient solution yet known for use by both classic and quantum computers (NIST, 2024). The standardized cryptographic algorithms available in this category include ML-KEM, a method to provide secure key exchange, and ML-DSA, a method to provide digital signatures compliant with FIPS 203 and FIPS 204, respectively (NIST, 2024).

Compared to Elliptic Curve Cryptography (ECC), lattice-based methods typically require larger public keys and ciphertexts to achieve equivalent levels of security; they therefore require greater amounts of memory and bandwidth to store and communicate them. However, because of their strength and practicality, these methods may be used in various applications including TLS, VPNs, Cloud Computing and IoT type devices (NIST, 2024).

2.2.2 Code-Based Cryptography

The security of code-based cryptography comes from the computationally hard task of decoding random linear error-correcting codes. The most well-known implementation of this concept is Classic McEliece, which has resisted all known attacks for over three decades and remains one of the top candidates for providing post-quantum security (NIST, 2023). Although it is highly secure, the use of code based cryptography is constrained due to extremely large public keys that increase the amount of memory needed for storage and reduce its ability to operate in low resource environments such as those found in many mobile devices and embedded systems (NIST, 2023).

2.2.3 Hash-Based Cryptography

Hash-based cryptography uses a secure cryptographic hash to generate digital signatures instead of depending on number theoretic problems. Thus, the security of hash based cryptography depends on the collision resistance properties of the hash used, which makes it currently the most conservative of all the approaches in post-quantum cryptography (NIST, 2024). SLH-DSA is the current standardized hash-based digital signature scheme from FIPS 205 and was originally called SPHINCS+ (NIST, 2024) and although this approach provides excellent long-term security for signatures generated with it, it also generally produces much larger digital signatures and longer times to perform signing operations compared to ECC-based digital signature schemes.

2.2.4 Multivariate Cryptography

A number of cryptographic proposals based on solving systems of multivariate polynomials are currently being researched as possible post-quantum solutions (NIST, 2023) due to their ability to potentially resist attacks from both classical and quantum-based computers. As such, while these schemes were originally thought to be secure, they have since been shown to be insecure by means of attacks via cryptanalytic methods. Due to this, many researchers now view the multivariate families with a lower level of trust than was previously the case; thus, additional study will need to take place prior to being implemented in practice.

2.2.5 Isogeny-Based Cryptography

Isogeny based cryptography offers a new way to use elliptic curve concepts by using the mappings of elliptic curves through the process of isogenies. The initial reason for interest was due to smaller than typical key size with potential resistance to attacks via quantum computers (NIST, 2023). Recent results have led to compromise of multiple top isogeny based cryptographic schemes. Therefore, there has been an immediate loss of faith in the method and limited involvement of isogeny-based cryptography in current post-quantum standardization activities (NIST, 2023).

2.3 Hybrid ECC-PQC Approaches

Hybrid cryptography uses conventional algorithms such as ECC and post-quantum cryptographic algorithms to protect users from current and potential attacks. As opposed to switching directly to post-quantum cryptographic algorithms, the two types of algorithms can be used together during the transitional phase. For example, a TLS connection could use ECDHE and ML-KEM to produce a common session key through a combination of contribution from both classical and post-quantum mechanisms (Stebila et al., 2025).

One of the most significant advantages that hybrid cryptography offers is an increase in cryptographic resiliency. When weaknesses in one type of algorithm are found, the other will continue to provide protection, which will reduce the total amount of security risk. The hybrid layer design makes it possible for organizations to transition to new cryptographic technology at their own pace, while still protecting their existing infrastructure (Stebila et al., 2025).

Hybrid deployments are also feasible due to the fact that many of the existing technologies being utilized today, including TLS, VPN's, cloud services, financial systems, and enterprise authentication platforms, rely heavily on ECC. Changing these existing

infrastructures would necessitate substantial modifications to existing software, hardware security modules, certificate management systems, communication protocols, and regulatory compliance procedures. Using a hybrid model will allow organizations to start introducing post-quantum protection into their systems without causing disruption to their current operations (NIST, 2023).

Although hybrid cryptography has several benefits, there are some drawbacks. Hybrid models will create increased computational overhead by running both ECC and PQC concurrently. Increased processing time, memory utilization, larger keys and larger amounts of communications bandwidth are all examples of what could occur when running both algorithms at the same time. Vendors may have varying hybrid implementations, so compatibility issues may exist if vendors do not support the same post-quantum algorithms. Organizations need to consider how they will address performance and inter-operability needs when considering a hybrid deployment model (Stebila et al., 2025).

In summary, hybrid ECC-PQC creates a viable path for organizations to move toward quantum-resistant security. This method will enable organizations to improve their cryptographic defenses today while allowing them to transition slowly toward fully post-quantum cryptographic systems (Stebila et al., 2025; NIST, 2023).

3.0 Literature Review

3.1 Paper 1: Hardware Deployment of Hybrid PQC (Azarderakhsh et al., 2021)

Research Objective: This study focuses on the transition requirements of quantum security systems and verifies the feasibility of a hybrid security model that integrates classic ECDH with post-quantum cryptography to support this transition.

Methodology: The researchers relied on FPGA hardware to build an ECC-PQC hybrid key exchange system, and conducted performance evaluations across latency, hardware footprint, and communication overhead.

Main Findings: The hybrid system integrates classical and post-quantum cryptography and only incur moderate overhead. It can be deployed in embedded resource-constrained environments. However, SIKE used in the system carries a risk of being cracked, reducing long-term validity. The study demonstrates that ECC does not require immediate full replacement and can coexist with PQC.

Advantages: Practical FPGA solution, Low power and efficient design and Verifiable hybrid architecture

Limitations: SIKE later broken, Limited evaluation scope and Security and compatibility concerns

Relevance to Project: This paper demonstrates hybrid ECC-PQC systems as transitional solutions before full post-quantum migration.

3.2 Paper 2: Post Quantum Cryptography Review (Bavdekar et al., 2022)

Research Objective: To analyze quantum computing impact on cryptographic systems and review post-quantum algorithms.

Methodology: Comprehensive literature review covering PQC algorithms, quantum attack models, and NIST standardization work.

Main Findings: ECC and RSA are vulnerable to Shor's algorithm. Lattice-based cryptography is identified as a strong candidate for post-quantum security. ECC relies on computational hardness, which quantum computing can break.

Advantages: Strong theoretical coverage, Includes multiple PQC categories and Up-to-date with NIST work

Limitations: No experimental implementation and Theoretical review only

Relevance: Provides theoretical justification for replacing ECC in quantum environments.

3.3 Paper 3: Post-Quantum Readiness and Migration (Gupta & Mittal, 2026)

Research Objective: To develop a structured migration framework for transitioning from ECC to post-quantum cryptography.

Methodology: Uses risk modelling, NIST standards, and migration planning frameworks.

Main Findings: Organizations must begin migration immediately due to "Harvest Now, Decrypt Later" threats. Hybrid ECC-PQC systems are recommended during transition. Migration is necessary as encrypted data may be exposed in future quantum computing scenarios.

Advantages: Practical enterprise framework, Based on standards and Real-world applicable strategy

Limitations: Focus on cloud systems only and Limited benchmarking

Relevance: Supports practical ECC migration strategies in real systems.

3.4 Paper 4: Cryptographic Dependency Inventory (Lee et al., 2026)

Research Objective: To identify cryptographic dependencies in decentralized systems before migrating from ECC to PQC.

Methodology: Uses taxonomy-based framework to map cryptographic usage across systems.

Main Findings: ECC is deeply integrated across system layers, making replacement complex and requiring structured migration planning. ECC is part of a larger cryptographic ecosystem.

Advantages: Strong system-level analysis, Useful for blockchain systems and Structured migration mapping

Limitations: Limited implementation testing and Focused on decentralized systems

Relevance:

Highlights complexity of ECC replacement in real infrastructures.

3.5 Paper 5: Hybrid PQC Authentication for 6G (Turnip et al., 2026)

Research Objective: To evaluate hybrid ECC-PQC authentication systems for 6G networks.

Methodology: Comparative survey of hybrid authentication protocols and security performance.

Main Findings: Hybrid cryptography provides backward compatibility and quantum resistance. Future systems require both ECC and PQC integration.

Advantages: Future-oriented research, Covers 6G architecture and Strong hybrid focus

Limitations: Survey-based only and No experimental implementation

Relevance: Supports hybrid ECC-PQC as dominant transition architecture.

4.0 Critical Discussion

As per the reviewed literature, it can be observed that there is a general agreement in the reviewed literature that the basic security assumption behind ECC, computational hardness of the discrete logarithm problem, is fundamentally challenged by the progress being made in quantum algorithms.

4.1 The Quantum Threat to ECC-Based System

The five papers that have been reviewed all show that the Elliptic Curve Discrete Logarithm Problem (ECDLP) is the only potential weakness in ECC. For a mathematical proof that Shor's algorithm makes 256-bit ECC impractical, read the paper by Bavdekar et al., 2022. But Azarderakhsh et al. (2021) show that this is not enough: Even if ECC is encased in a hybrid wrapper, the ECC primitive is still the "weakest link. This is the problem that if a cryptographically relevant quantum computer (CRQC) comes to life tomorrow, then hybrid systems are still broken: the adversary just attacks the ECC part of the key exchange.

In emerging infrastructures, turnip et al. 2026 extend this risk, stating that 6G authentication protocols that are currently being standardized will need to abandon ECC altogether to prevent it from being hacked. Unlike Gupta & Mittal (2026), the immediate threat for an enterprise system that already exists is not that it will be decrypted by a CRQC (which

is not yet in existence), but rather the "Harvest Now, Decrypt Later" strategy. This is an important difference because while Azarderakhsh speaks of the technical vulnerability, Gupta & Mittal speak of the operational vulnerability, thus meaning that organisations need to take action today whether or not the quantum computer is 10 years away.

4.2 Current Challenges in Replacing ECC with PQC

Paper 1 and Paper 4 are in contradiction with each other. From a hardware point of view, the challenge posed by Azarderakhsh et al. (2021) is whether it is possible to include PQC in an FPGA? (They showed us that it's possible, with reasonable overhead).

Instead, Lee et al. (2026) consider the challenge from a systems architecture perspective. They believe that this is very easy to do with a hardware device, but the challenge lies in the cryptographic dependency web. ECC is not limited to TLS libraries, but also embedded in blockchain smart contracts, hardware security modules (HSMs), and legacy certificate authorities. Lee's taxonomy-driven approach shows that the process of replacing ECC is not only a matter of code but a rewiring of governance and consensus systems of decentralized systems. This means that there is a need for domain specific migration strategies, because otherwise the domain-specific approach of a one size fits all (as suggested by Gupta & Mittal) will not work in a decentralized environment.

Furthermore, Bavdekar et al. (2022) call attention to the issue of standardization. NIST has come a long way, but many PQC schemes are only just being subjected to cryptanalysis, and have not seen the sort of decades of testing that has been applied to ECC. The recent cryptanalysis advances for SIKE that were used in the hybrid implementation in Azarderakhsh et al. (2021) clearly show this risk. An algorithm that is deemed to be promising in the standardization phase might be broken, leading to loss of trust in migration strategies relying on certain PQC primitives.

4.3 Advantages and Disadvantages of Hybrid ECC-PQC

All papers confirm that Hybrid ECC-PQC is the most realistic interim option, but several papers offer a contrasting perspective on the pros and cons of this approach. Hybrid systems offer "dual protection" as shown by Azarderakhsh et al. (2021) and recommended by Gupta & Mittal (2026). In case the PQC algorithm is vulnerable to quantum attacks, the ECC layer is protected; and in case the ECC layer is vulnerable to quantum attacks, the PQC layer is protected. The "belt and suspenders" approach reduces migration threat.

Overhead (Performance vs. Security): Azarderakhsh et al. (2021) measure the overhead and find it to be acceptable. But Bavdekar et al. (2022) point out that these percentages are misleading; if these microservices are being processed by a million machines per second then this overhead becomes huge, making ECC as popular as it is in IoT seems to be a contradiction.

Most important is the critical-thinking point that is the failure of SIKE. Azarderakhsh et al. (2021) selected SIKE for its "smaller keys". Soon after their publication, SIKE was severely broken by classical attacks. This risk was anticipated by Bavdekar et al. (2022) who identified isogeny-based cryptography as being "risky" and "under-researched. This is an important lesson: If the underlying math doesn't work, then there's no point in hardware feasibility. Hence, the isogeny-based hybrids do not need to be considered by organizations and the finalized lattice-based standards (ML-KEM/DSA) of NIST should be followed.

4.4 Current Research Trends and Future Directions

The literature reveals a clear shift in the question from "Can quantum break ECC?". (Bavdekar) "How can we replace it without breaking the internet? (Lee & Gupta). The most progressive trend is being "quantum safe by design," as described by Turnip et al. (2026). They propose 5G/6G networks should be designed from the ground up to be capable of crypto-agility where algorithms can be changed by software patches without hardware changes as retrofitting would be inefficient. Additionally, Lee et al. (2026) note that automated cryptographic discovery tools are the next frontier; organizations simply don't know where all their ECC keys are located.

4.4 Critical Evaluation and Recommendations

Based on a critical synthesis of the reviewed literature, the following actionable insights are derived. Prioritize Inventory over Implementation. Even before implementing any PQC, organisations need to perform cryptographic dependency inventory as described by Lee et al. (2026). When replacing ECC without understanding how to integrate it, it can be more of a risk to break the critical infrastructure.

As shown by the SIKE break of the NIST-FIPS standard schemes (ML-KEM, ML-DSA), organizations should stick to NIST-FIPS standardized schemes and avoid trying to use smaller keys of less mature schemes.

Use hybrid TLS with IETF-standardized hybrid X25519MLKEM768 (Stebila et al., 2025) to ensure compatibility while providing quantum resistance. Harvest Now, Decrypt Later threat is existential for data that needs 20+ years of confidentiality they should not go through the hybrid phase and should go directly to pure PQC.

5.0 Conclusion and Future Work

5.1 Summary of Key Findings

This project will analyse five research papers that are recent which would be a critical investigation of the effect of quantum computer on Elliptic Curve Cryptography (ECC). The

results clearly demonstrate that ECC's security backbones the Elliptic Curve Discrete Logarithm Problem (ECDLP) are vulnerable to Shor's algorithm in the case of a large enough quantum computer. The "Harvest Now, Decrypt Later" threat is an operational risk to long-term confidential data, but large-scale fault tolerant quantum computers are not yet available.

Results show that hybrid ECC-PQC systems are the most viable solution in the short term from a migration perspective, providing cryptographic security and being backwards compatible while transitioning to PQC. Yet important hurdles remain, such as the highly deepened integration of ECC at multiple layers of the system, performance overheads in resource-limited settings, and the possibility of cryptanalytic breakthrough on unproven PQC schemes like the catastrophic failure of SIKE soon after its deployment in hardware was publicized.

5.2 Future Research Directions

Based on the gaps identified in the critical discussion, the following future research directions are recommended. The authors of Lee et al. (2026) point out that automated cryptographic discovery tools are a future direction that requires further research to discover and map cryptographic dependencies in complex, decentralized systems. Manual inventories simply are not possible on a large scale.

Performance optimization for constrained Devices, Azarderakhsh et al. (2021) showed that it can be implemented on FPGA, but more research is needed to optimize PQC algorithms for ultra-low-power devices such as IoT/edge devices where bandwidth and battery life are highly constrained.

Long-term cryptanalysis of lattice schemes, continued cryptanalysis is needed to ensure the survivability of ML-KEM and ML-DSA against both classical and new quantum attack vectors over the next 20-30 years, even though they are NIST-standardized.

Future protocols, especially those for 6G, recommended by Turnip et al. (2026) should be designed to be crypto-agile in the sense that cryptographic primitives can be replaced with software instead of hardware or infrastructure changes. Research into consensus mechanisms that are quantum-safe, not just more secure than ECC, is essential for blockchain and decentralized finance (DeFi) systems, where ECC is woven into the very fabric of the network's governance.

Finally, the quantum threat to ECC is serious and is very close; however, there is a structured, phasing approach that can be followed with hybrid architectures and NIST-standardized PQC algorithms that can lead to quantum-safe cryptographic infrastructures. By starting their inventories and migration efforts now, organizations will be best prepared to reduce risks from future quantum adversaries.

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APPENDICES

Minutes of Meeting

Project: Impact of Quantum Computing on Elliptic Curve Cryptography (ECC)

Meeting: 1

Date: 23 June 2026

Time: 10.35 A.M.

Location: Whatsapp group

Attendees:

- Keertennah Devi A/P Ponnambalam
- Shwetha A/P C.Ravichandran
- Netrha Shri A/P Rama Moorthy
- Haress Sivakumar

Agenda / Discussion:

1. Discussed and divided the sections of the report among group members.
2. Agreed on a timeline for the first draft.

Action Items:

Task	Assigned To
Draft Introduction	Netrha
Draft Background	Shwetha
Draft Literature Review	Haress
Critical Discussion and Conclusion	Keertennah

Minutes of Meeting

Project: Impact of Quantum Computing on Elliptic Curve Cryptography (ECC)

Meeting: 2

Date: 27 June 2026

Time: 2.00 P.M.

Location: Whatsapp group

Attendees:

- Keertennah Devi A/P Ponnambalam
- Shwetha A/P C.Ravichandran
- Netrha Shri A/P Rama Moorthy
- Haress Sivakumar

Agenda / Discussion:

1. All members confirmed that they had completed their respective draft sections. The drafts were shared in the group for review and compilation. Agreed on a timeline for the first draft.
2. Shwetha create the initial PowerPoint slide deck. She shared a link with the group so that all members could add their specific points.
3. Keertennah took responsibility for compiling all individual drafts into a single, cohesive report document, ensuring consistent formatting, numbering, and flow
4. The group discussed and finalizing the report and recording the presentation video. It was agreed that all members would review the compiled report and add their presentation slides allowing one final day for recording and submission.

Action Items:

Task	Assigned To
Submit completed draft sections	All members
Create PowerPoint presentation and share editing link	Shwetha
Add content/slides for respective sections	All members
Compile and format the final report	Keertennah
Review final report for errors	All members
Record video presentation	All members
Final submission of PDF with video link	All members

Next Meeting:

No further meeting scheduled.